Client prepares shares offline · five independent sharekeeper deposits · recovery sheet stays with the client

Legend: Client / offline · Sharekeeper custody · Integrity / sheet · Validation

CLIENT — TRUSTED OFFLINE DEVICE & RECOVERY SHEET

1. Generate shares

Create five SLIP39 shares in one group with a 3-of-5 member threshold.

2. Encrypt each share

Each share gets its own independent random 100-character one-time pad. Never reuse pad material.

3. Hash ciphertexts

SHA-256 of each exact ciphertext. Hashes stay only on the recovery sheet.

4. Build recovery sheet

Five pads, five hashes, sharekeeper contacts, local record numbers, encoding, and version.

5. Local dry-run before relying

Test reconstruction offline. Any SLIP39 passphrase stays on the client device.

SHAREKEEPER CUSTODY — FIVE INDEPENDENT TITLE COMPANIES (numbered 1–5; no shared directory)

Enroll separately (live video or optional in-person) → deposit that company's ciphertext + admin metadata only

Sharekeeper 1

One ciphertext only
+ admin metadata
Own local record ID

Sharekeeper 2

One ciphertext only
+ admin metadata
Own local record ID

Sharekeeper 3

One ciphertext only
+ admin metadata
Own local record ID

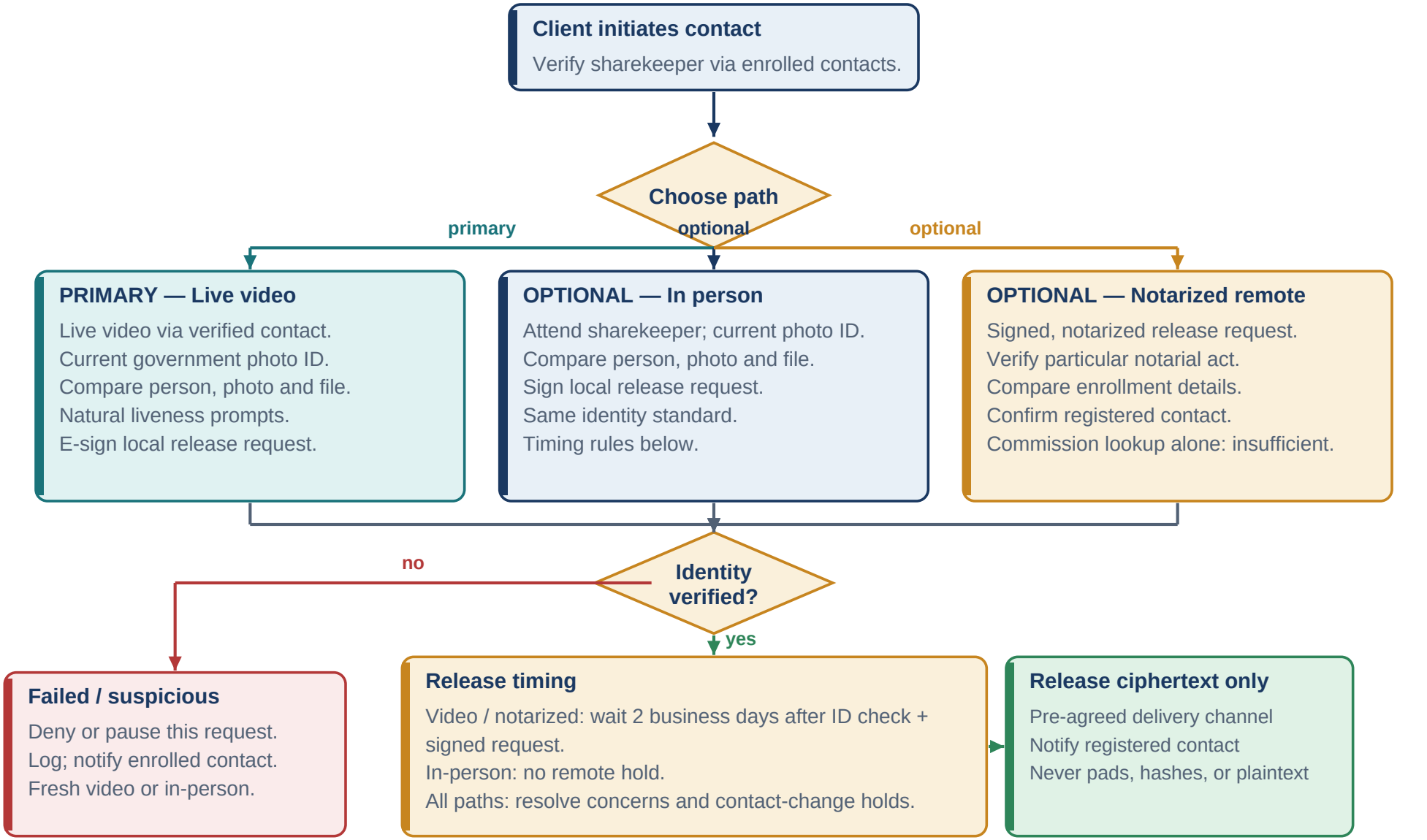
Sharekeeper 4

One ciphertext only
+ admin metadata
Own local record ID

Sharekeeper 5

One ciphertext only
+ admin metadata
Own local record ID

Never at sharekeeper: pads, plaintext shares, recovery sheet, client-supplied hashes, reconstructed secret, SLIP39 passphrase, or co-sharekeeper identities. No sheet-possession proof.

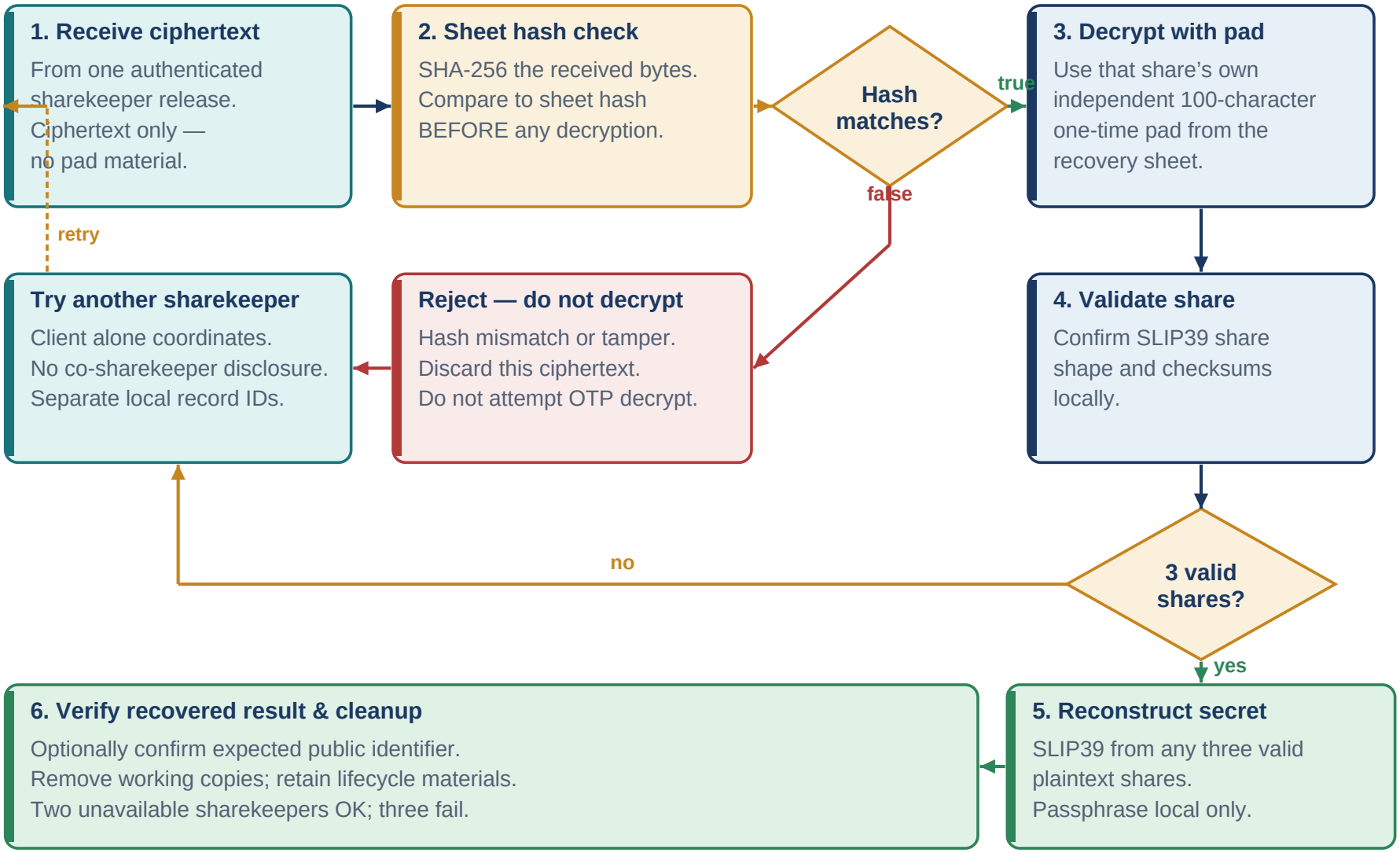


Contact change (separate hold)

Same identity standard (video or in person). Notify old channels when possible.
TWO BUSINESS DAY change hold. Do not release on new details until ALL holds expire.
Lost phone or email: fresh video or optional in person — no hard lock.

Hash check before one-time-pad decryption · client alone counts to three · sharekeeper never sees pads or sheet hashes

Trust boundary: CLIENT offline device + recovery sheet | SHAREKEEPER holds ciphertext only (separate local record IDs)



Client-held ciphertext hashes are for local integrity only — not a possession proof shown to sharekeeper. Sharekeeper can hash its own stored ciphertext independently; the sheet hash is never supplied to sharekeeper.

Availability vs confidentiality · sheet is the primary secret · rotation replaces the full material set

Lost recovery sheet

Without a separately prearranged pad backup, recovery is impossible even with all five ciphertexts. An optional sealed duplicate improves availability but creates another sensitive target.

Compromised sharekeeper

A correct one-time pad keeps plaintext confidential if one ciphertext leaks. The authentic sheet hash detects ciphertext substitution. The company can still refuse or destroy its copy (availability risk).

Stolen sheet + three ciphertexts

An attacker who has the sheet and any three ciphertexts can reconstruct without passing sharekeeper authentication. Sharekeeper procedures cannot revoke already-leaked ciphertexts.

Incapacitation or death

A representative may verify by live video or optional in-person attendance, and must separately establish legally valid authority. Pad access needs a separate estate arrangement. Authority cannot recreate missing pads.

Rotation / refresh

Generate a complete fresh share set: new pads, ciphertexts, hashes, and a versioned sheet. Test before retiring the old set. Never reuse pads or mix versions. Old retained materials still reconstruct the old secret — move assets if exposed.

Local-device compromise

Neither video nor in-person authentication protects decryption or reconstruction on malware-infected hardware. The authentic sheet and a trusted local recovery device remain essential trust assumptions.

Video and in-person share the same identity-confidence / release standard; that is not a claim of identical fraud resistance. Sharekeepers must never store pads, plaintext, sheet hashes, secrets, passphrases, or co-sharekeeper directories.